THE BEST OF 2600

[A HACKER ODYSSEY]

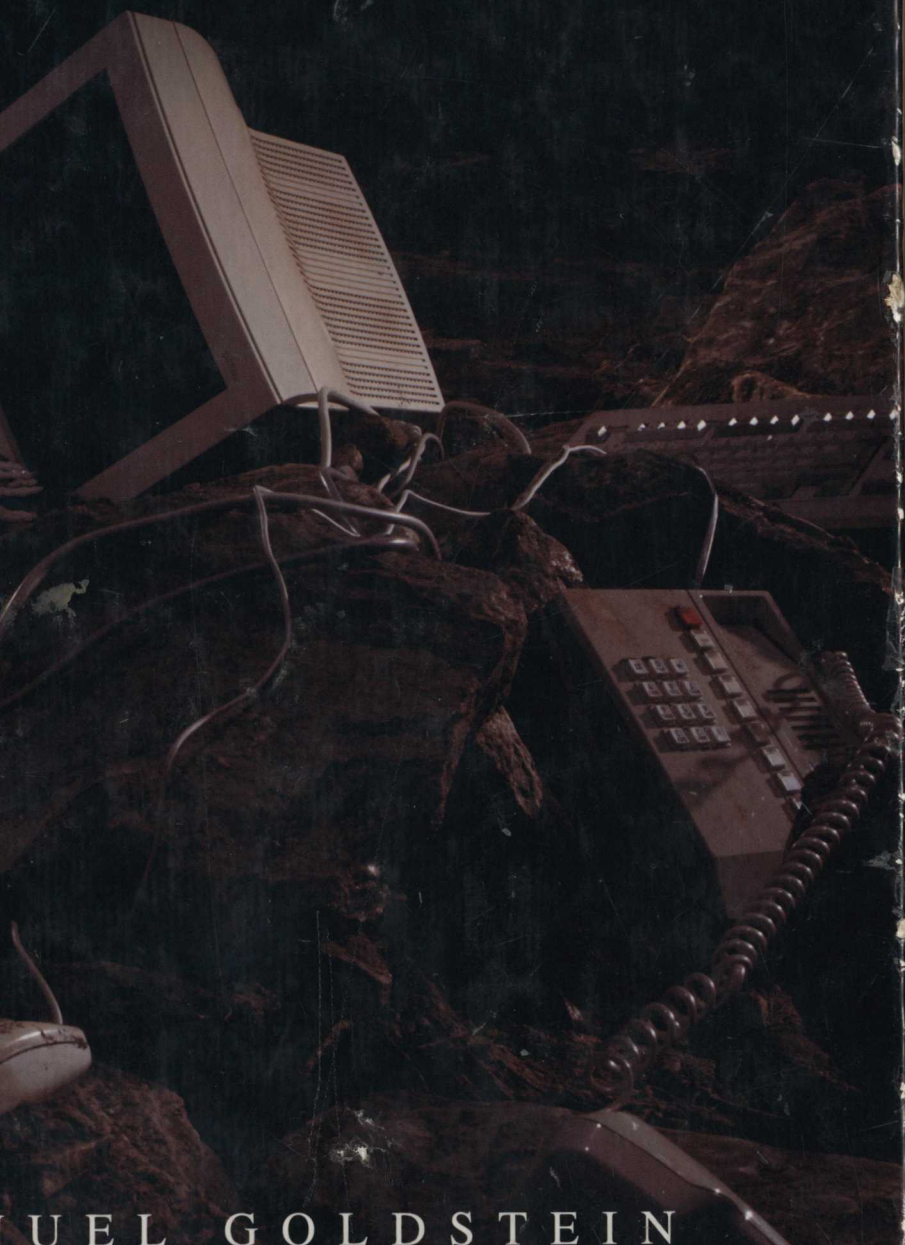


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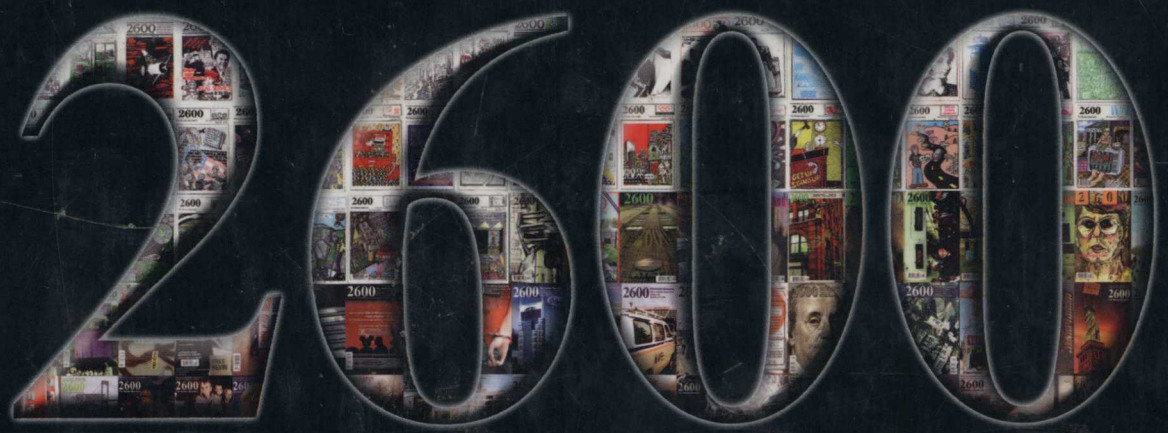
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24 years of enduring exploits, creative controversy, and hackers who made history

It's colorful, controversial, cutting-edge — and you can't wait to read the next issue. Since its birth in 1984, *2600: The Hacker Quarterly* has published the discoveries and adventures of hackers worldwide. Now you can enjoy the best of them all in this entertaining, provocative collection. From the first curious and intrepid souls who discovered they could outwit Ma Bell to those who've hacked the Department of Defense and ParadisePoker.com, they're all here, telling their stories in their own words.

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- The world vs. Kevin Mitnick
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- Simplex locks — so simple to break
- A hacker in Iraq
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- Hacking Google Adwords

Emmanuel Goldstein has been editor and publisher of *2600* from the start. He also hosts a weekly radio show, *Off the Hook*; served as technical consultant for the 1995 feature film *Hackers*; and has testified before Congress on the issue of hacking and where the threat actually lies. You can visit the *2600* web site at www.2600.com, or contact *2600* by mail at *2600 Magazine*, PO Box 752, Middle Island, NY 11953.

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About the Author

Emmanuel Goldstein (emmanuel@goldste.in) has been publishing *2600 Magazine, The Hacker Quarterly,* since 1984. He traces his hacker roots to his high school days in the late '70s, when he first played with a distant computer over high-speed, 300-baud phone lines. It didn't take long for him to get into trouble by figuring out how to access something he wasn't supposed to access. He continued playing with various machines in his college days at the State University of New York at Stony Brook. This resulted in an FBI raid as he once again gained access to something he really shouldn't have. It was in the midst of all of this excitement that he cofounded *2600 Magazine*, an outlet for hacker stories and tutorials from all over the world. The rapid growth and success of the magazine was both shocking and scary to Goldstein, who to this day has never taken a course in computers. Since 1988, he has also hosted *Off The Hook*, a hacker-themed technology talk show on WBAI 99.5 FM in New York City. In addition to making the hacker documentary *Freedom Downtime*, Goldstein hosts the Hackers On Planet Earth (HOPE) conferences in New York City every two years, drawing thousands of hackers from all over the world.

You can contact *2600* online at www.2600.com or by writing to *2600 Magazine*, P.O. Box 752, Middle Island, NY, 11953.

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Acknowledgments

This is far and away the toughest part of the book to write. How do I sum up in words what so many people have meant over so many years? I'll undoubtedly forget someone, they'll notice, words will be exchanged, and new enemies for life will be formed. It's a painful yet necessary part of the process.

From the beginning there have been people who have been there to encourage me in various endeavors, *2600* included. And there have also been those who have actively sought to discourage me and painstakingly point out the many errors of my ways. I would like to thank each of these groups equally. The former gave me the positive reinforcement that helped convince me that this was all worth it and that it would eventually have a beneficial and lasting effect. The latter gave me the obstinacy and unmitigated wrath to prove them wrong. A positive outlook fueled by anger is really all you need to succeed in this world.

On to specifics. First, I must thank three people who helped me wade through well more than 1,000 articles to sort out the ones that would eventually appear here: Tony Fannin, Mike McTeague, and Kevin Reilly. It was a mountain of material, but I managed to scale it successfully thanks to their help. The many people at Wiley who dealt with our rather unconventional way of piecing together a book deserve particular thanks, especially Carol Long and Maureen Spears. The hard work of my agent, Cameron McClure, made this all come together with remarkable speed and clarity.

My cofounder, Dave Ruderman, deserves special gratitude for coming up with the name "2600"—which I initially hated—way back in 1983. Those late nights of plotting and scheming are among my most favorite memories on this planet. My good friend Dave Buchwald defies any sort of description. He's done everything from office management to cover design and is somehow able to come up with new skills overnight like some sort of power computer. We've also been blessed with some truly talented layout artists over the years—Ben Sherman, Scott Skinner, Terrence McGuckin, and Mark Silverberg—all of whom I'm honored to call my friends. And without Mary Nixdorf, our office manager, *2600* would last about a day. Her incredible dedication and attention to detail make the whole enterprise function smoothly—a feat that people to this day tell me is impossible. Our previous office managers (Pete Kang and Fran Westbrook) also got us through some vital periods of our growth. People like Mike Castleman, Carl Shapiro, Mike Kaegler, Ed Cummings, Rob Nixdorf, Nick Jarecki, Kevin Mitnick, and Mark Abene have always been there to offer encouragement, expertise, and words of advice for all sorts of projects over the years, no matter how crazy they may have actually been.

And some of our projects have been pretty bizarre. And, speaking of bizarre, the magazine covers, T-shirt art, web site work, and various designs for the HOPE conferences, put together by such artists as Frederic Guimont, Holly Kaufman Spruch, Kiratoy, Kerry Zero, Tish Valter, and Affra Gibbs, have been nothing short of eye-opening in every regard.

Oh, yes, the HOPE conferences—how could I forget? Since 1994, we've been holding hacker conferences in New York City called Hackers On Planet Earth, which have drawn thousands of people from all over the globe to the historic Hotel Pennsylvania. And this has all been made possible by a phenomenal volunteer effort, which would take many more pages than I have to adequately acknowledge. Nothing symbolizes the power of the hacker community more than seeing hundreds of people come together to pull off a feat like this every couple of years, a feat which is, of course, impossible. Having these conferences is indeed a real motivation to keep doing what we've been doing. It's one thing to sit back and write stuff from some remote location; it's quite another to actually meet the audience and hear their stories and realize that it all actually matters.

And none of this would have ever been possible (for me, at least) without the creative inspiration that I got from working at a magical place called WUSB, the noncommercial radio station at the college I went to: the State University of New York at Stony Brook. My fondest memories of the station include cohosting the eclectic program "The Voice of Long Island" with my good friend Mike Yuhas back in the early '80s. That place (both the college and the station) taught me so much about diversity and imagination. I was able to steer all of the creative energy from there to the various projects that I became involved in after graduation. I can honestly say that none of this would have ever happened were it not for that initial inspiration. This also led to my involvement with another magical place: WBAI-FM in New York City, a full-power noncommercial station that reaches four states, accepts no corporate money and basically exists to challenge the status quo. (Do I even have to point out how impossible this is as well?) They gave us an outlet for the hacker perspective on technology and *Off The Hook* was thus born. That radio show gave hackers a voice and served as a valuable staging ground for everything from the Free Kevin movement to our defense against the Motion Picture Association of America when they decided to sue us. Particular thanks must go to past program director Andrew Phillips, who recognized this need before even I did.

Three of my very best friends in the world—Walter, Naftali, and Huey—have given me the spirit I needed when I needed it the most. Inspiration also came from family, especially Patricia O'Flanagan, who taught me to think for myself, and Monica Clare, who always makes me remember the value of the small things in life. There are many other relatives who I must also thank for just letting me do my thing without trying too hard to stop me.

Thanks must also go to Mike Tsvitis who printed our magazine for decades and truly helped us achieve levels of excellence we had only dreamed about.

The international community of hackers continues to inspire me every day. Nothing is cooler than going to some far off remote land and discovering that there are people there who are asking the same questions, performing the same experiments, and generally engaging in the same level of mischief as those of us back home; it's just further proof that hacking is a distinct part of the human genome.

And of course, none of any of this would have happened were it not for the many people who have written for *2600* over the years. Even if your work doesn't appear in these pages—for that matter, even if your work hasn't been published in the magazine itself—know that your interest, feedback, and willingness to actually put pen to paper and come up with something different and engaging is the driving force for everything that we've been doing since 1984. I cannot thank all of you enough.

And finally of course, a special thank you to God for not striking me down with a bolt of lightning despite the many requests s/he must get on a daily basis. Respect.

Introduction

The question we get asked more than any other at *2600* is how in the world did we pull this off? Meaning, I suppose, how did we manage to not only put out a magazine for nearly a quarter century that was written by hackers but also to get the mainstream public to take an interest in our subject matter?

Well, it certainly wasn't easy. I guess the first thing to make clear—and probably the one fact that both those who love us and those who hate us can agree upon—is that it was never supposed to get this big. When we first started out in 1984, we never envisioned it going beyond a few dozen people tied together in a closely knit circle of conspiracy and mischief. Those first issues were three sheets of paper with loose leaf holes punched in the sides stuffed into envelopes. In late 1983, we sent messages to a bunch of bulletin board systems (BBSes) that had hacker content on them. In these messages, we invited people to send in self-addressed stamped envelopes and in return they would get a free copy of the premiere edition of our new hacker magazine. I'll never forget the thrill I got from seeing the first responses come in the mail.

As for content, we had grown into an interesting group of storytellers and educators by way of the BBSes. Basically, by logging onto one of these systems, we would be able to find other people who seemed able to string together a sentence or two and either tell an interesting tale of a hacker adventure or explain to someone exactly how a particular computer or phone system worked. This is how the core staff developed. And we always knew there would be more people out there to add to the mix.

For me, this was a natural expression of my various interests fueled by all sorts of inspirations. Computers had fascinated me ever since I first encountered one in my senior year of high school back in 1977. I never was a programmer, and to this day I have never taken any sort of computer course. That would have taken all of the fun out of it. No, for me the computer was the ultimate toy, a device that could spit back all sorts of responses and which had almost endless potential. My main interests, though, were writing and media. I came from a family of writers and my major in college was English, plus I had worked in some capacity on every high school and college publication I encountered. Then there was my involvement in radio. I was lucky enough to stumble upon WUSB at Stony Brook University, a freeform noncommercial radio station where I was encouraged to be creative and alternative in all sorts of different ways. So when you added all of these elements together, the volatile mix that was to become *2600* seemed almost inevitable.

When we mailed that first issue, we didn't know what to expect. Arrest and impris onment was one possibility that crossed our minds. There was, after all, an investiga tion underway into some of the people involved in the magazine before we had eve published our first issue—something to do with logging onto computers that didn belong to them using other people's names. Back then, having a computer was some thing reserved for very few people. There was no Internet to explore. Apart from th BBSes, the only way to learn about real systems that actually did something was to fig ure out a way to get connected to them and absorb as much as you could. It was neve about being malicious or destructive, although even then we had our hands full fight ing that misconception, which was fueled by the mass media. We were a diverse bunc of curious folk, exploring a new universe and sharing our findings with anyone wh cared to listen. We were dangerous.

Strange things started to happen after we sent out the first *2600* in mid January o 1984. People started to send in checks for an entire year! Our magazine became th talk of the BBS world and, we would later learn, numerous corporate boardrooms an government agencies. It seemed such a simple idea and yet nobody else was doing i There had been a newsletter before *2600* known as *TAP*, which had started publishin back in the '70s with the help of Abbie Hoffman and a bunch of Yippies. It was a fu publication but it came out sporadically and eventually stopped altogether in the earl '80s. What people saw in *2600* was something previously unheard of in this commu nity: consistency. Every month at the same time we released a new issue. And not onl was it consistent, but it actually looked somewhat professional, thanks to my recentl acquired job as a typesetter for an unsuspecting local newspaper. It really felt as everything had come together at just the right time for *2600* to be born.

After the first year, when people started to actually *renew* their subscriptions, w knew we were on to something. The word continued to spread, more writers came ou of the woodwork, and the media followed our every move with rapt attention. Whil technology was booming, it was still very early in the whole computer revolution. W were seen as pioneers, and I quickly became a "computer expert" even though I ha never taken a course and wasn't particularly technical. It didn't seem to matter. An time something happened involving computers or telephones, it was assumed the sta of *2600* knew all about it—that is, if we weren't in fact accused of being responsible fo it in the first place!

We expanded from six pages (three double-sided sheets of paper) to eight page (two really big sheets folded in half) and kept that format until 1987 when we decide to try something new entirely. We became a magazine in the true sense of the wor with a color cover and staples and a total of 24 pages. But the workload and expens for this kind of a format quickly began to exceed our resources, so we switched to quarterly format in 1988 with 48 pages. Shortly after that final format change, we go on the radar of magazine distributors and began to see *2600* show up at newsstand

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change my mind. "2600" summed it all up. It was all about reaching out and grabbing technology, making it do what you wanted to do, and communicating with people all around the globe. Not to mention the fact that in any alphabetical list of publications, we would always be first. It was a match made in heaven.

Of course, running the magazine itself has been anything *but* heaven. When you deal with material that is—to put it mildly—controversial, you wind up with an impressive number of powerful people who want to see you go down in flames. Our very existence has embarrassed almost every major corporation at some point, resulted in numerous emergency board meetings, and made some people's jobs a bit harder. None of that was our intent, although that's little comfort to those affected. What we've always been primarily interested in doing is simply getting the information out there and watching it grow into something productive. Phone companies have learned not to leave sensitive billing information on computers with default passwords that anyone can access. Credit agencies now actually work to protect all of that data they keep on every one of us. And the people who design secure systems, many of them our readers and sometimes writers, know how to think like hackers, which makes their creations innovative and flexible. I believe we've contributed quite a bit of good to the world of technology and things are better now than they would have been had we never come on the scene. Of course, that doesn't mean there haven't been numerous attempts to put us out of other people's misery. But when someone believes firmly in freedom of speech and full disclosure, it's kind of impossible to shut him up.

What has amazed me the most in the decades that followed is that the interest level has never subsided. Over the years, more and more people have become entranced not only with the technology itself but also with its social implications and overall importance to the future of humanity. It may sound a bit heavy handed but all of this—the development of the Internet, computers being used as printing presses, the prevalence of low-cost or free telecommunications all around the world, the sharing of information and resources—is having a profound impact on the human race in ways that no one from our forefathers to Aristotle could ever have predicted. Somehow we wound up right in the middle of all the turmoil. And just like it felt back in the early days when everything just sort of came together at a particular moment, this feels like the right people are in the right place at the right time to test the system, develop new tools, and keep freedom of speech alive.

A Story of Eavesdropping (April, 1986)

Everybody knows an old man who was in the Second World War and has plenty of war stories to tell. Well sometimes it pays to take the time to listen.

We knew that the enemy was monitoring all of our international radiotelephone channels, despite the sophisticated voice-scramblers which "inverted" speech, making high tones into low ones and vice versa. Only authorized persons were permitted to use overseas telephone circuits.

We were equipped with elaborate recorders and switching control boxes which permitted us to cut off either side of a conversation, or to substitute ourselves for either party. A strict set of rules forbade us to permit maritime information, weather reports, cargo information, etc. to pass over the circuits.

Influences in Washington sometimes resulted in orders issued to us to permit use of the overseas telephone circuits, even though we were suspicious of previous conversations because parables and unusual phrases often used, made it difficult to follow what was being said. "How can we monitor carefully, when we can't understand what they're saying?" went unheeded.

We caught one fellow red-handed in South America using weird terms like "birds leaving the nest with a basket of eggs." I finally cut in the circuit and told him I'd forgotten what they meant. He tried a couple of other phrases, which I also couldn't understand. Finally, he lost his patience and blurted out, "Oh hell. I'm talking about those special munition orders which left yesterday for Germany."

By this time, a special telephone speech scrambler had been developed which was small enough to fit and use on a desk. Its availability was extremely limited, but a couple of army officers—one in the U.S. and the other in Panama—had been able to get hold of a pair of them, and between them secretly installed them on their desks, unbeknownst to us of course!

One day I heard the fellow in Panama say, "OK Joe, now over to the scrambler," and their ensuing conversation became unintelligible. We quickly checked the radio telephone circuit equipment and discovered that the technical characteristics of the equipment they were using and our own were identical. As a result, when they inserted their scramblers the speech inversion righted itself and their conversations went out over the radiotelephone circuit in clear language—readable by anyone! That was the end of the use of their private "secret conversation system."

Some of the worst offenders of overseas telephone use security were the top people. I'll have to list Generals Eisenhower and Marshall as two of them—at least sometimes. I can remember one day the circuit between London and Washington happened to be very poor in quality and "understandability" was stretched to the utmost.

General Marshall in Washington had General Eisenhower on the line in London who couldn't understand a word of what Marshall was saying. Marshall repeated several times "Ike, this is GCM—Marshall—GCM—got it?" without results. Finally in frustration Marshall turned to an aide and could be plainly heard to say "What's the code word for my name?"

The next thing we knew, Marshall was slowly and distinctly repeating his code name interspersed with "GCM" and "Marshall." Of course, we had to cut the circuit and notify the code group in Washington to immediately "bust" the code—we couldn't take any chances—revelation of the code word for his name might have been all the enemy intelligence was waiting for to help it "code-break" other communications.

On the other hand, President Roosevelt and Prime Minister Churchill were two of the best and easiest to monitor. Both used references to previously transmitted overheard messages by numbers and most of the conversations were along the lines: "Well Winnie, on number 528, I really don't think we should do that—you know how they are." Nobody could gain any information from listening to their telephone conversations.

I always enjoyed listening to Sir Winston originating a call. The British telephone operators were required on *every* connection to announce in advance of a conversation: "You are warned not to mention the names of vessels, sailing dates or conditions, cargoes, weather, etc., etc., etc., any violation on your part will result in the circuit being cut off and your action being reported to the highest authority. Do you understand?" Sir Winston always dociley replied. "Yes ma'am. I understand."

One enemy group had learned the "language" of speech inversion. For example, listening on the air to a radiotelephone circuit, one might hear a word that sounded exactly like *krinkanope*. That was the word *telephone* after it had passed through the speech inversion system!

The First Atomic Bomb: A True Tale (March, 1984)

This story was originally related by Laura Fermi, widow of the nuclear physicist Enrico Fermi who, along with assorted colleagues, participated in the first test bomb in the desert outside Alamogordo, New Mexico, in the early morning hours of a summer day in July 1945.

When the date had been established for the secret test, staff members from the Manhattan Project (as the secret test was known) were invited to bring their spouses to New Mexico to watch the results of the several years of research. Each staff member had been assigned specific tasks to handle while there. Generally, they acted as observers and were stationed in a circle around the perimeter of the bomb site. Enrico and Laura were stationed in an area about twenty miles to the southwest of the bomb site.

The morning came when the bomb was scheduled to be detonated in the test. Laura told it like this…

Enrico and I woke up at 3:00 a.m., to go to the site. The test was scheduled for 4:30 a.m. that day, which was July 19, 1945. We drove to our post, about twenty miles from the site. It had been arranged that the nearly 100 of us present would be located in a circle about 100 miles in circumference surrounding the bomb site. We were all to be in communication with each other over telephones, all of which were connected through the exchange in Alamogordo.

We arrived at the site at 4:15 a.m. and almost immediately it began to rain, quite a heavy, very typical torrential downpour during the summer. We waited in our car, and at 4:30 a.m. the time came and went, but the bomb did not go off. Enrico and I assumed it might have been postponed due to the rainstorm, but decided to check with the other staff members to see for sure. For some reason, the telephone there at the site did not seem to work; the operator would not respond. *(Note: At that time, nearly all phones in the United States, and certainly in New Mexico, were manual. No dialing of any sort was possible—you had to use the operator for everything.)*

Finally Enrico decided that we would drive into town and try to contact the others and see what went wrong. So we drove back to town, and got there about 5:15 a.m. The only place open at that time of night was a hotel, and we stopped in there to use a pay phone. Strangely enough, the pay phone was not working either, or at least the operator never came on the line to ask what we wanted. Enrico was quite curious about all this and decided to investigate. We went outside the hotel, and Enrico found where the telephone wires came off the pole and down into the building. He decided that we would follow the wires, so we walked down the street looking overhead at the wires on the pole as we went along. Finally, we turned down one street and saw a house. The telephone poles and wires from all directions seemed to come down to this house. There must have been hundreds of wires from telephone poles all coming down onto the side of this house and going in through an opening.

We noticed that there was a front porch light, which was on. The front door was open, but there was a screen door, which was closed. We went up on the front porch and looked into the house. A switchboard was there, and there were a dozen or more lights on the switchboard lit, blinking off and on as people were flashing the switch hooks on their phones trying to raise the operator. The room was just dimly lit, and near the switchboard was a sofa, and a woman was laying on the sofa sound asleep! Enrico pounded very loudly on the screen door, and shouted at the woman. Suddenly she opened her eyes and looked at him, very startled. Then she looked at the switchboard. Immediately she sprang up, dashed over to the board, sat down and began frantically answering the calls.

Without saying any more, Enrico and I left, went back to the hotel where our car was parked, and drove back to our monitoring post twenty miles out into the desert. We had been at our post only about five minutes when the explosion went off, at about

6:30 a.m., which was two hours behind schedule. Later, we talked to the other staff members and found that there had been some confusion because of the rain. None of them had been able to reach the others because the telephone operator had fallen asleep, and the phones were not getting answered/connected…

We on the staff all had a big laugh out of it, but nothing more was ever said or done, and I doubt to this day that that woman is even aware that the first atomic explosion in the world was delayed two hours because of her.

Amazing, but true. Alamogordo was a tiny town back in the '40s, and it's very doubtful that the night operator had ever seen so much traffic in her life as the hundred or so people all on the line at once that early morning. More than likely, the poor dear had had a very rough day the day before, in the miserable summer heat, had been unable to sleep during the day, and had come to work that night thoroughly exhausted. She probably decided that "it won't hurt just to close my eyes for a minute…," and the rest of the story is already told. After all, experience had taught her that in fact she would not usually get a dozen calls all night on her shift, and she felt relatively safe in stretching out "just for a minute."

Numbers That Led to Trouble

Having access to telephones and the increasing variety of new uses for them invariably led to all sorts of fun for the average phone phreak. (It was quite common in the hacker world to append a "ph" to any word with an "f" if it had anything at all to do with phones, such as "phreak," "phriend," or "phraud." Sometimes the exact opposite was true, which resulted in the word "fone" popping up every now and then.) Of course, fun in the hacker world usually translated to trouble in the real world, which pretty much summed up where hackers fit in societywise. Our innocence and adventure was always seen as evil and threatening to the uninitiated who couldn't seem to understand what motivated these strange individuals to go and play with phones for hours on end. As a result of our hard work, we would share whatever strange phone numbers we were able to discover with anyone who was interested in hearing about them. Because there weren't as many methods of communication as there are today and because there were significantly less phone numbers floating around, discoveries like the ones that follow seemed to mean a lot more. And then, of course, there was one of the all-time favorite phone phreak pastimes: running a teleconference. The unparalleled pleasure of hooking several dozen of your closest friends together and going on a telephonic voyage around the world was something so few people could even conceive of in the 1980s. And yet, *everyone* in the hacker community had some sort of encounter with teleconferences back then. Some were even rumored to have spent most of the decade connected to one.